

THEORY AND PRACTICE IN COMPOSITION

BY HANS GÁL

MUSICAL theory as a basis for practical composition is a problem as old as music itself, and one obviously without any definite solution. Conditions change, and theory has to be readjusted accordingly. The theory of harmony has slowly and cautiously developed to account for the more recent harmonic phenomena. Counterpoint, as a rule, is still based on the old method that seems to have been in practical use centuries before it was first codified by J.J. Fux in his 'Gradus ad Parnassum' of 1725. We try to adapt this method to our use, and there is a great deal of argument and difference of opinion about it. I confess that I have never found a text-book I could use for the teaching of counterpoint or composition, or even of harmony, and I am afraid those of my colleagues who do not take these things as a mere matter of routine have the same story to tell.

No one could penetrate to the very core of this problem more thoroughly and earnestly than Paul Hindemith has done in the preface to his 'Unterweisung im Tonsatz' (¹) I can only re-emphasize every sentiment of this genuine musician's profession of faith which shows the keenest awareness of the momentous importance of a sound technique for the future of his art. It may thus seem contradictory that, being in full accord with Hindemith's attitude as a whole, I nevertheless disagree with every single detail of his theory. If a musician and a composer of such distinction develops a new theory, one is bound to take it very seriously; but having gone through his book with an increasing feeling of uneasiness, I regret to state that to my mind Hindemith's theory, consistent as it is, is the logical result of a basic error—an error the more dangerous because it amounts to a truism. This is one reason why I feel impelled to contradict it; my other reason is that the point in question seems to me to be of the utmost importance if we are to find a way out of the present disorder.

It is necessary to go down to the very root of the whole problem. Theory has two different aims: to give a plausible and consistent explanation of existing musical phenomena and to furnish a method for teaching the technique of composition. Our accustomed theory of harmony is largely devoted to the first, more or less analytical purpose, counterpoint mainly to the second. But in addition both these branches of theory have a very important educational design. Their study is still the only universally accepted method of forming and developing a mental ability which is the most decisive factor for a musician: musical imagination, mental conception and reproduction of harmony and polyphony. This applies not only to the composer, but to the performer as well, if he lays claim to any standard, and it especially applies to the conductor.

To Hindemith's mind it is an indisputable fact that vocal music having been deposed from its reigning position as long as three centuries ago and superseded by instrumental music, has definitely to be abandoned as a basis for musical theory. The instrumental conception of music, as represented by Bach, Beethoven or Wagner, brought about a gigantic development by which the whole outlook on the art has been changed altogether. Nevertheless counterpoint, as it is generally taught to-day, continues to be based on the old, obsolete notion of vocal conditions. An absurd situation no doubt! Still more absurd, if we realize that the diatonic scale, on which counterpoint is still solidly and rigidly founded, tends more and more to adopt the semitones of the chromatic scale as ordinary material in general use. According to Hindemith, the diatonic scale as a basis of tonality was definitely overthrown in Wagner's 'Tristan'; the full disintegration of tonality followed as an unavoidable result. From this Hindemith concludes that the only consistent and practical solution would be to adopt the chromatic scale of the

tempered system as a new basis for tonality. Schoenberg discussed the same idea in his text book on harmony as early as 1911 and founded on it his Zwölftontheorie. The whole argument sounds extraordinarily obvious and irrefutable; all the same it is a false conclusion, based on an error from the outset, as I am prepared to prove.

Hindemith has devoted a large part of his book to research into the tonal material, examining and verifying the intervals and their natural derivation. His system, based on the natural harmonics, does not work, however, without some artificial readjustments. I do not criticize it, only I think that it would be difficult to find any tonal system the naturalness of which could not be equally well proved, if proofs of that kind were conclusive. . The point is that music is primarily not a material but a psychic phenomenon. What matters is not the interval, the harmony, the succession of sounds; the question is “How does our ear react to that interval, that harmony, that succession of sounds?” Or, to put it still more precisely, “How does our mind, consciously or unconsciously, interpret those acoustical phenomena?” As long as we perceive only single sounds, there is no music. The sequence of sounds has to be organized into an intelligible pattern before we can receive the impression of music. This necessary organization of sound relates to harmony—the sum of simultaneous sounds—as well as to melody—a coherent phrase of rhythmically disposed notes. Polyphonic music calls for a still more complex kind of perception, different phrases resounding simultaneously in various rhythmical combinations. We shall have to examine this complicated psychic act and its conditions, if we want to understand the nature of music.

There are three reactions to music by which it can be experienced as an actual event: invention (composition), performance and listening. Let us observe the progress of a phrase, as the simplest musical unit, from its origin to its final destination—the listener’s ear. The composer, writing it down, has a clear idea of it in his mind. He may have previously shaped and reshaped it, played it on his piano, improvised on it; but in every phase of his act of invention he must have had a clear, precise mental conception, otherwise he would not deserve to be called a composer. What is a mental conception of a phrase? It is an image of it, an inner awareness, as clear as the conception of a word or a sentence, an image enabling the composer to reproduce it by singing or whistling. He could not imagine an interval, a phrase, a melody without the potential ability to reproduce it. Our musical imagination, however, is not confined to the narrow compass of our actual voice. We have no difficulty, for example, in imagining the high violins of the ‘Lohengrin’ Prelude, or the sinister double basses at the entrance of Othello in the last act of Verdi’s opera, or even a theme extended over four octaves like the opening of Richard Strauss’s ‘Heldenleben’, if its single phrases are clearly cut and jointed. A trained ear has no difficulty either in imagining harmony, and harmonic as well as melodic perception is closely connected with the potential ability of reproduction. When imagining a harmony, I feel every note of it as if I were singing it, and I am able to sing in my mind simultaneously all parts of a polyphonic ensemble. Thus mental conception of music is an exact equivalent of vocal reproduction.

Let us proceed to a psychological analysis of the act of performing music. A conductor, lifting his baton for the first beat, must have a clear image of what is going to happen. He has studied his score, or if not, should be able to read it as you read your newspaper, comprehending at a glance the meaning of a phrase or a harmony, and being perpetually ahead of events by the imaginative power of his musical intellect. Otherwise he would not be capable of leading his musicians, his first and most important job being to control and ensure the correctness of what is being played. To control something implies the capacity of confronting it with an ideal mental image. This inner conception, if correct, is precisely same as the composer’s, and formed under exactly the same conditions: the pattern has to stand so vividly before the conductor’s mental hearing that he would be able to reproduce it vocally. I know perfectly well that there are conductors whose ability to do this is rudimentary. I have seen

many who, with but vaguest notion of what was to come, could beat time with amazing self-confidence, and I remember the following naïve confession of a colleague of mine: "You can never know in advance what you will hear when you read a modern score, can you?" Which did not prevent my honourable friend from advancing to the rank of Generalmusikdirektor in Germany. I have also known composers who had no more than a very dim notion of what they had written down, but were nevertheless taken quite seriously by conductors, critics and a defenceless public. Such phenomena are symptomatic of a great deal of confusion and decadence in our art and cannot be accepted as a counterproof to the above observations.

As a matter of fact I have found more unmusical individuals among conductors than among practical performers, instrumentalists or singers. They too have to "control" the sounds they produce by confronting them with an inner, preconceived image. It would be altogether impossible, for example, to play the violin in a mechanical way, simply by exact fingering and a well-trained left hand. The player would probably lose his pitch as soon as he ventured on a higher position, and apart from a definite lack of purity you would notice at once that he was playing without understanding, just as you would notice it if somebody were reading to you something, say a poem, without being aware of its sense. Just as speech does not consist of words, but of sentences, so does music not consist of intervals, but of phrases. Being aware of a musical phrase implies, for a player as well as for a conductor, the ability to preconceive it mentally or, which comes to much the same thing, the faculty of reproducing it vocally. This conclusion is not affected by the existence of bad performers who do not understand what they are playing, nor by the fact that there is a class of instruments on which indeed playing is possible without a controlling ear: the keyboard instruments. Theoretically speaking, a deaf person could play the piano; music, for him, could simply be a system of signs indicating how to move his hands in order to put the right fingers on the right keys.. I have met pianists who, without being deaf, had a marvellous faculty of eliminating their hearing and musical intellect when playing. But such playing, like a deaf person's performance, has nothing whatever to do with music.

As for the final reaper of all the toil and trouble involved in a musical performance—the listener—we do not speak of him, we hold him in contempt, and he has been so much abused, humiliated, neglected that he has become quite modest and meek. He will swallow a thistle and take it for asparagus, if told to do so with sufficient authority. But what is his genuine feeling? It is amazing how differently individuals react to a piece of music, and how variously the same listener may respond to the same piece of music at different times. There is a whole world between the passive listener, who would not even recognize a piece he has heard yesterday if played to him again next day, and the memorable achievement of Mozart when he wrote down from memory Allegri's 'Miserere' after having heard it only once. In this case an unfailing perceptive faculty was coupled with a miraculous musical memory. As regards our present topic, we are mainly interested in the first of these gifts. Experience shows that the highest degree of perceptive capacity, a recipient working with the precision of a phonograph, is rare even among musicians. The recipient, more-over, will not always work with the same reliability; this may vary according to actual disposition, attentiveness, concentration. So, on the average, musical perception will be more or less incomplete.

I wish to stress one point: I only refer to a brisk, conscious musical perception as a result of active, concentrated listening. To cultivate this musical education, knowledge of theory or analysis is by no means indispensable. But neither would a passive, dreaming attitude do. Try to memorize a poem during a musical performance or to solve a crossword puzzle, or simply switch off your mental collaboration: the result will be the same, the music will not penetrate into your consciousness. I should never accept as musical perception a process of absorbing passively, like a kind of hashish. There may be a certain sensual charm, for a very short time, in

the sensation of a kind of musical kaleidoscope, of an iridescent acoustical dust without any shape or consistency; but if you have received an impression of that kind, the music, to my mind, has been no music, or you have not got hold of its proper contents. In either case your subjective impression is no infallible criterion. Startling dissensions of opinion, such as have occurred when famous masterpieces were performed for the first time, are due less to differences in taste than to differences in perceptive faculty. Any musician will have experienced that his opinion may change by closer acquaintance with a piece of music he flatly rejected when he first heard it.

This understood, it will be asked in what consists the act of musical perception, as the main condition of genuine musical appreciation? Let us take, once more, the simplest unit of organic music, a phrase. It has a definite meaning, just as a spoken sentence has a meaning which is entirely and exclusively musical, quite apart from its emotional value, which, as a secondary psychic phenomenon, does not concern our present research. The sense of a musical phrase is determined, explained, emphasized by other phrases and by the way they are interlocked. If you lose the thread of this continuous flow, you are lost yourself; your impression becomes chaotic. To perceive a phrase, to experience it as an actual, intelligible fact, is exactly the same as to understand the sense of a spoken sentence, where you cannot have understood the speaker if you cannot repeat it. Whether you are capable of reproducing it ten minutes later depends on your memory; you will probably have forgotten the exact words, but you will remember the meaning. Musical perception works in exactly the same way. A phrase has not penetrated into your consciousness unless you are capable of reproducing it mentally. Your musical memory may not be sufficiently developed to retain it, but at the moment of its actual perception the phrase is a clear, definite fact to you. Thus musical listening involves a continuous shaping and reshaping of musical facts in your mind, an activity representing a receptive analogue to the composer's and the performer's work. "Understanding" of a musical phrase implies the capacity of reproducing it mentally, i.e. the potential ability to reproduce it vocally. Or, to put it the other way round: any music we are able to appreciate as a mental experience is vocally conceived; otherwise we could not make head or tail of it.

The result of this analysis seems to me convincing. It shows that any actual, vivid musical experience—not only the composer's and the performer's, but even the listener's—is conditioned by a vocal conception, and that, where this potential is missing, musical experience is bound to be incomplete and disconnected. If this fact is understood, the following conclusions become inescapable:

1. The human voice is the eternal, unshakable basis of music;
2. Instrumental music has been derived from, and is deeply rooted in, this soil. Its development as an independent branch will always be restricted by the natural limits of vocal conception.
3. Theory therefore can only be founded on vocal conditions.

I should not like to be misunderstood. I have not the slightest inclination to renounce the glorious achievements of instrumental music in the last three centuries, nor to sacrifice anything of value conquered in the course of this development. But we must realize what matters and where possible danger lurks. Do not let us forget this : vocal imagination goes a long way. There are, however, parts in Beethoven's 'Missa solemnis' which I never found satisfactory, even in the best performance, and which will remain a problem for ever, simply because they are written against the human voice. Yet I would not miss a bar of that beloved masterpiece, nor would I miss any of Bach's glorious unvocal, choral fugues, as we find them in the Mass in B minor and everywhere in his cantatas and motets. Only do not advise anybody to imitate that style, nor tell me that a new way of development has been opened by such violations of the natural character of a material, because they have been committed by a genius. We should distinguish between the absolute value of a masterpiece and its usefulness as a

model and guide. I would, as a pianist, rather forgo the complete works of Chopin, beautiful though they are, than a single one of Beethoven's last sonatas, with all their problems of keyboard style. I am fully aware that not one bar could be altered without playing havoc; nevertheless I do not see a way for development along that line of writing for the piano, and as a matter of fact nobody has ever ventured on it.

When instruments took the lead in the second half of the seventeenth century, all sorts of musical forms emerged, founded to a greater or lesser degree on instrumental principles. But at the same time opera developed, and with it came the most lovely achievement of the Italian genius for music—cantabile melody. Instrumental music was substantially influenced by it, and it remained one of the most efficient determinants of formal development throughout the classical and romantic periods. Here is one of the sound, indestructible roots connecting instrumental music with its original soil. The slow movement of a symphony, the second subject of a sonata-form work, the trio of a scherzo—they are genuine singing episodes, vocally conceived. This explains even the naïve barbarism of adapting words to a popular melody taken from a classical masterpiece, the kind of thing done already in Beethoven's time, with the *Allegretto* of his seventh symphony, for example, and experienced with a shudder in our own time with the incredible popularity of Schubert's misused melodies. Such tunes are always essentially vocal, and it is this which explains their powerful effect on the listener's imagination. The dualism of motion, the instrumental, and *cantabile*, the vocal principle, has remained one of the fundamental agents of formal development. Motion in itself alone, the type of "perpetuum mobile", is only bearable for a very short time. If it goes on and on, the listener will have a feeling of unrest and discomfort, and he will stop listening.

The practical question is this: what is essentially vocal? To answer it we have to bear in mind some constitutional peculiarities of the human voice, all of which are indivisibly connected with fundamental attributes of our musical imagination. Our voice is confined to a certain compass; it has a definite, though not unconquerable objection to diminished and augmented intervals as well as to major sevenths and intervals greater than an octave; and it needs stability in an interval to reproduce it. The position of all the steps of the diatonic scale is clear and unequivocal. The position of an F# in F major, for example, may be quite clear and obvious too. But if the function of an interval is doubtful, if its position, owing to harmonic or melodic conditions, wavers between F# and G flat, it becomes unsteady, with the result that our mental perception the interval becomes uncertain and the intonation is endangered. To put it simply : chromatic intervals are difficult, enharmonics disastrous. As I have pointed out above, the actual compass of our voice does not coincide with the limits of our vocal imagination, but an over-tension by several consecutive huge intervals in the same direction, especially if they are of an unharmonious kind, goes against the grain. Who, I wonder, would be able to sing the following example of Hindemith's?



I know it is not meant to be sung ; but it is meant to be understood by a pupil who, as the first

condition of musical integrity, must learn to control his inner hearing. If he is sincere, he will be altogether discouraged by a model of this kind; if he settles down at his piano in order to invent such stuff with the aid of that expedient, he will become a humbug. The Hindemith example quoted above, by the way, is undoubtedly diatonic, if we understand as diatonic a scale of seven degrees each of which may be altered by an accidental. As old-fashioned theory of harmony found out ages ago, a chromatic passing-note or an alteration into an accidental leading-note does not affect the essential meaning of a step or a harmony. I have not yet been able to find any chromatic situation in 'Tristan' which could not be explained in that way. There is not the slightest proof for Hindemith's assertion that "in Wagner's 'Tristan' the rule of the major and minor key has been broken. Instead of the diatonic scale the chromatic scale has been substituted here as the basis of melody and harmony." Well, there can be no doubt in any bar of 'Tristan' to which major or minor key it belongs. If Wagner had invented a new tonality for 'Tristan', he would not have returned so deliberately to the old tonality, as he did in his next work, 'The Mastersingers', with a more pronounced diatonic texture than he ever found before. Schoenberg gave a more plausible explanation when he stated that Wagner's abundant use of certain chords liable to enharmonic change, such as the chords of the diminished seventh' and the augmented fifth, fascinated his followers and gradually undermined the feeling for tonality in the subsequent period. This fact is indisputable; the only question is whether development in this direction was progress or a calamity. It is likely that Wagnerism has done more harm to the music of all nations than any other "-ism" at any time. We are recovering slowly from that illness, and the developments of the last twenty years seem to me to indicate that, in spite of all the apparent confusions, we have passed the turning-point. Hindemith himself, as a practical musician, realized the danger of chromaticism when he wrote :

If the harmonic conditions become too difficult to survey, or progressions follow each other in such a way that their meaning is not unequivocally clear, our ear becomes uncertain. The player or singer does not realize the position of the "comma" [the difference of a quarter-tone between enharmonic notes], and his intonation becomes impure. Therefore combinations founded on pronounced chromatics or enharmonics are very difficult to perform. For choristers they are frequently altogether impracticable, even after all the experience the musical ear has undergone in history. Practice of music, composition and theory, must never forget the postulates established by the fact of pure intervals and the requirement of our ear to find them reaffirmed as frequently as possible.

I have nothing to add to this comprehensive statement but an explanation about the comma. Hindemith's acknowledgment of its existence implies a feeling for intervals which is incompatible with an acceptance of the chromatic scale as a basis. As long as you feel the comma, as long as you want to distinguish between an F# and a G flat—as every musician will do, if his ears are not hopelessly corrupted by the piano—your musical conception is based on the diatonic system. To feel the comma implies referring the note definitely to the higher or the lower step of the diatonic scale, or to bring it into the orbit of the diatonic system. The chromatic scale is sexless, without a beginning or an end, and without the possibility of a comma, which would upset the whole order of twelve equal steps. To search for a tonic in the chromatic system would be as hopeless as to look for the end of a ring or the start of eternity. The same applies to the whole-tone scale, and for the same reason. Schoenberg was more consistent when he deposed the tonic once and for all. Hindemith takes infinite trouble in combining opposite and incompatible principles, and the result is a deplorable muddle. One fact stands to reason: our ear, our musical imagination, is unable to realize indeterminate, diatonically unrelated chromatics. That seems to be one of its constitutional attributes, since it

is confirmed by other musical systems which have developed independently. The Chinese, the Indian, the Arabic systems are based on diatonic scales. Chromatic intervals and quarter-tones, as far as they are used there, are confined to sliding melismata, which correspond exactly to our “passing notes” and do not alter the original pentatonic or heptatonic basis. In the diatonic system the difference of whole tones and semitones decides the character of a step. Without the possibility of relating a sound to a definite degree of the diatonic scale our ear loses its orientation, as it definitely does in an undecided enharmonic situation.

At the beginning of this century the old diatonic system had gone the way of disintegration. But we must not forget that infinite variants of diatonic scales are possible, that the major and minor tonality and the medieval modes represent only a small fraction of diatonic possibilities. My personal belief is that we shall have to go back to the source, to find a new genuine vocal style and to make an end of dishonesty and humbug, as they have spread increasingly, since the piano began to influence music in a poisonous way. It is a doubtful advantage for a composer to replace consistent, concentrated mental work by more or less random, uncontrolled improvising. I know a composer who, he told me, gets his nicest inspirations by striking a wrong key when improvising. There is another danger in the new world of sound, opened up by the pianoforte pedal. Is it ever realized how orchestration during the last hundred years has been influenced by the piano? This would be an interesting subject for a special examination. Look for example at the opening of Strauss’s ‘*Alpensymphonie*’, where a B flat minor scale descends slowly through three octaves, every step of it held by a fraction of the muted, divided strings. To this soft humming sound, used as a background, a theme is intoned by the low brass instruments. No human ear is able to hear through this diffuse noise, a successful imitation of the worst use a pianist could possibly make of the pedal. I know the intention is to depict the dim outlines of a huge mountain seen through the early morning mist; but unfortunately the mountain is made of pasteboard and the mist is a smoke-screen, put up in order to hide the fact that there is no music behind it. The pedal has introduced indefinite, inarticulate sound into music. I call it dishonesty, but that may be a matter of opinion.

The history of music shows that development does not proceed in a straight line, but rather in a spiral. It looks as if at the beginning of this century instrumental music had arrived at the climax of a development leading to the very edge of a precipice. Beyond is chaos, a kind of music which no sincere person will pretend to grasp, if grasping means a clear, controlled perception of musical facts. I had the clearest possible confirmation of this when Rudolf Kolisch, whom we all know to be a passionate apostle of modern music, told me after his first performance of Webern’s string Trio: “We had more than fifty rehearsals, but we don’t understand it yet.” When nobody but the composer himself understands a piece of music, a stage of development is reached where there is no practical difference between subjectivism and madness. The author may be quite honest and sincere; but so is the lunatic who pretends that he has five arms and seven legs. In literature and the fine arts, where similar, symptoms of hyper-individualism came forth some thirty years ago, the illness seems to have been overcome by now. In music the situation still appears rather confused. It is impossible, of course, to define where the “natural limits” are to be found. It stands to reason that they are fluid. Musical hearing has changed and developed a good deal during the last four centuries. Only one must not over-rate the speed of that change: it may take as many centuries, while a change of style may only take decades. We have no right to assume that, as far as our primary capacity of musical perception goes, we are born with ears different from those of our grandfathers. The development noticeable in the public appreciation of modern music—which is very slow, anyhow—is a result of individual education, not of changed conditions of original hearing, and it is confined to a small minority of educated people. No wonder that to the vast majority, even in the most progressive countries, modern music is as strange as Chinese. This attitude is perfectly explained by the theory I tried to develop before now. I remember an old uncle of

mine bluntly declaring music to be worthless if he could not whistle it. His musical horizon was rather narrow and did not extend beyond musical comedy and popular operatic melodies. We made fun of him. But at least he was perfectly sincere, and we must realize the abyss between the average capacity of primary musical hearing and the exigencies of serious contemporary music. Here is one of the main difficulties we are confronted with in finding a practical method for the teaching of theory to a young musician, whose ear and whose mind have to be opened to musical phenomena of the widest range. The only way I have found practicable is to resort to counterpoint, the strict counterpoint as determined by the style of the sixteenth century. I am no advocate for Fux's rules and regulations, nor for Cherubini's or Bellermann's. The rules of strict counterpoint, as I apply them, are in direct relation to the music of Palestrina and his contemporaries. Anyhow, we have to get down to conditions of sufficient clearness and simplicity to enable the pupil to become aware with impeccable lucidity of what he hears in his imagination. Many years of teaching experience and innumerable experiments as to method have not yet shown me a better basis than the Palestrina style. Starting from this point, we have to develop the pupil's imaginative capacity and sense of polyphony step by step. He has to go mentally through all the stages of development, as a human embryo does physically. It takes time and trouble, but there is no way of dodging these. The risk is, of course, that he may find the method tedious, but I have always held such teaching to be successful if after two years of serious studies a pupil comes to the conclusion that his talent is not sufficient. Another question is how to avoid aridity and dullness in a study which is comparable only with methodical technical exercise on an instrument. The answer is: combine it with living music. A pupil who regularly attends choral rehearsals and performances, listening, singing, accompanying and conducting alternately, and who has experienced such masters as Palestrina, Lassus, Marenzio, Byrd or Morley as makers of living, beautiful music, will write his counterpoint exercises with quite a different feeling than he would if they were merely so much paper work to him. Moreover, he will never be in danger of forgetting what vocal setting means. Let him study and practise Bach's fugues when he is working up fugue: if he is gifted, he will profit more by this experience than by any theory. Let him make practical acquaintance with several instruments, give him a chance of doing chamber music, coaching singers, taking part in orchestral rehearsals. That is the only way to educate a musician and to keep his mind free from the dust and mould of paper and speculation. You cannot give him a ready-made technique, because that does not exist and never has existed. You can only develop in him a sound basis for its acquisition by his own indefatigable work, a basis founded on a vivid sense of style and formal balance, and on an inexorable artistic sincerity. There is only one text-book for this, and it is a huge one: the masterpieces of four centuries. But make no mistake—only a master can use it.

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¹ Part I. (Schott, Mainz & London, 1937).